William Huang

Electrical Engineering Student | Hardware, Systems & Circuit Design
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PROFESSIONAL SUMMARY

Electrical Engineering student with hands-on experience in PCB design, embedded systems, hardware prototyping, and system integration. Experienced in designing custom electronics using KiCad, integrating microcontrollers and sensors, and developing embedded firmware in C/C++. Passionate about building practical hardware projects and creating innovative electronics solutions through rapid prototyping, testing, and iterative design.

TECHNICAL SKILLS & TOOLS

Hardware & Design: PCB Design (KiCad), Analog & Digital Circuits, Power Systems, Voltage Regulation, System Integration
Embedded Systems: Microcontrollers, Embedded C/C++, Firmware Development, Real-Time Systems
Interfaces: I2C, SPI, UART, USB
Tools & Software: MATLAB, Simulink, LTSpice, PSpice, Verilog, MPLAB X, Git (basic)
Lab Equipment: Oscilloscopes, Multimeters (DMM), Power Supplies
Other: Troubleshooting, Hardware Debugging, Data Analysis, English (Fluent), Mandarin (Fluent)

EXPERIENCE

Cal Poly Pomona - Bronco Star | BLADE, Team 1 — **Pomona, CA, September 2025 - May 2026**
Power Systems Subteam Lead

- Led electrical design and integration for a 1U BalloonSat (100,000 ft operation), including power distribution and sensor interfaces
- Designed and validated a 7.4V Li-ion power system with voltage regulation and protection circuitry
- Troubleshot system-level hardware issues using oscilloscopes, DMMs, and power supplies, resolving integration and performance faults
- Performed system-level integration of sensors, power subsystems, and embedded hardware across cross-functional teams
- Developed and refined a high-altitude system from concept to functional prototype in a fast-paced, iterative environment

PROJECTS

Post-Quantum Secure Telemetry for CubeSats (NASA ORBIT) — Verilog Hardware Design — **Pomona, CA, Feb. 2026-Present**

- Collaborated within a team to design and integrate Kyber cryptographic hardware into a secure telemetry system
- Designed modular arithmetic hardware (mod 3329) for post-quantum cryptography (Kyber)
- Implemented Barrett reduction and Montgomery multiplication for optimized performance
- Built self-checking testbenches with randomized inputs for verification
- Analyzed latency and critical path timing for performance optimization

Smartwatch Embedded Hardware Project (Kicad Self-Project) — **Pomona, CA, December 2025**

- Designed a complete ESP32-C3 embedded hardware system integrating power management, IMU sensors, and display interfaces.
- Created a multi-layer PCB layout with optimized component placement, routing, decoupling, and power distribution.
- Integrated I2C, SPI, and USB interfaces at the schematic and layout level.
- Verified component footprints, symbols, and pin mappings using datasheets and engineering documentation.
- Applied practical PCB design principles for manufacturability, assembly, and hardware validation.

Real-Time Fan Control System — Embedded Hardware & Control (PIC18F4620) — **Pomona, CA, November 2025**

- Designed a real-time embedded system integrating temperature sensing, PWM fan control, and RPM feedback
- Implemented a state-machine with manual/auto modes, dynamically adjusting duty cycle based on temperature thresholds
- Interfaced SPI (TFT), I²C sensors, ADC (photoresistor), and GPIO inputs for full system integration
- Developed interrupt-driven control using timers/counters and validated performance through debugging and testing
- Measured and monitored fan RPM and duty cycle behavior to evaluate system performance
- Developed low-level firmware in C for microcontroller-based real-time control using interrupts, timers, and PWM

EDUCATION

California State Polytechnic University Pomona — **Pomona, CA**
Bachelor of Science, Major in Electrical Engineering, — **Expected May 2027**
Cumulative GPA: 3.35
Relevant Coursework:

- **Electrical Engineering Fundamentals:** Electromagnetism and Circuits, Advanced Circuit Analysis, Digital Logic Design, Microelectronic Devices and Circuits, Microcontrollers, Signals and Systems, Power Engineering

INVOLVEMENTS

Clubs: IEEE Student Member (CES), Bronco Star/Space (BLADE)